

## PHASE 1: DATA INGESTION



### Wikidata Ingestion

item: Q154952 (Willem-Alex.)  
parent: Q159609 (Prince Claus)  
gender: male  
article: wiki/Willem-Alexander



### Wikipedia Article

**URL:** wiki/Willem-Alexander  
“Willem-Alexander was born... during the reign of Queen Juliana, as the eldest child of Princess Beatrix and Prince Claus...”

## PHASE 2: TRANSFORMATION & ENRICHMENT



### Convert to Turtle

```
data:Willem_Alexander a :Person ;  
  :hasParent data:Prince_Claus ;  
  :hasSex data:male .  
data:Prince_Claus a :Person ;  
  :hasSex data:male .
```



### Extract Raw Text

**S1:** Willem-Alexander was born... child of Beatrix and Prince Claus.  
**S2:** He became Prince of Orange upon his mother's accession.



### SWRL Inference

```
data:Willem_Alexander a :Person ,  
  :Man ;  
  :hasSex data:male ;  
  :hasParent data:Prince_Claus ;  
  :hasFather data:Prince_Claus .  
data:Prince_Claus a :Person ,  
  :Man , :Ancestor ;  
  :hasSex data:male ;  
  :isFatherOf data:Willem_Alexander ;  
  :hasSon data:Willem_Alexander .
```

## PHASE 4: TARGET DATASET PAIRS



### Text-Graph Aligned Output

#### Denoised Text:

“Willem-Alexander was born in Utrecht during the reign of his maternal grandmother, Queen Juliana, as the eldest child of Princess Beatrix and Prince Claus.”



#### Bound Gold Triples:

```
data:Willem_Alexander a :Man ;  
  :hasFather data:Prince_Claus .  
data:Prince_Claus a :Man , :Ancestor ;  
  :isFatherOf data:Willem_Alexander ;  
  :hasSon data:Willem_Alexander .
```

## PHASE 3: FUZZY MATCH ENGINE



### Fuzzy Match & Filter Logic

```
foreach sentence in raw_text:  
  if sentence contains alias in (Willem, Claus):  
    → Keep Sentence  
  else:  
    → Drop Sentence
```



**Sentence S2  
dropped**